SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Dry Run Evaluation

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 2 – Dry Run Evaluation
Weightage:	35% of CIA	**Duration / Marks:**	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	3rd Year	**Date:**	10-10-2026

Code of Conduct

I K Mahesh Sai, 192412674 (Name / Reg No) certify that this submission is my original work and that I have *adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K Mahesh Sai

Instructions

- Read each question carefully; answer all four questions (25 marks each, 100 in total).
- Show every dry-run step: the initial queue, stack, priority queue or belief state, each iteration, your assumptions and a clear final answer, and finish with the interpretation asked for.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Dry Run Evaluation rubric: understanding of the search problem, state expansion and dry run execution, application of the search algorithm, accuracy of the solution path, and presentation and logical reasoning.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end). Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	**Q2**
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	**Q1**
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	**Q1**
4. The Nature of Environments	Task-environment properties and their classification	**Q1**
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	**Q1**
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	**Q2**
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	**Q2, Q4**
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	**Q3**
9. Uninformed Search Strategies	Dry runs of BFS, DFS and uniform-	**Q3**

CSA17 – ARTIFICIAL INTELLIGENCE

Assessment Tool 2 – Dry Run Evaluation: Complete Solutions

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The solutions below follow the question paper's stated conventions. Search traces show the frontier after each expansion and test the goal when a node is removed from the frontier.

QUESTION 1: AGENTS, ENVIRONMENTS AND RATIONALITY

(a) PEAS description

PEAS component	Description
Performance measure	+10 for each square cleaned by Suck when it was Dirty; −1 for each Left or Right movement over 6 time steps.
Environment	Two-square vacuum world, A and B; both start Dirty; dirt does not reappear.
Actuators	Suck, Right, Left, NoOp.
Sensors	Current location and status (Dirty/Clean) of the current square.

(b) Task-environment classification

Property	Classification	Reason
Observability	Partially observable	The agent sees only its location and the status of its current square.
Number of agents	Single-agent	Only the vacuum agent makes decisions.
Determinism	Deterministic	Each action has a predictable result.
Episodic / sequential	Sequential	Actions affect later location and cleaning opportunities.
Static / dynamic	Static	The world and dirt status do not change while the agent deliberates.
Discrete / continuous	Discrete	Locations, percepts, actions and states are finite/discrete.

(c) Dry run for both agents

Step	X percept	X action	Y percept	Y action	Y known-clean squares after step
1	[A, Dirty]	Suck	[A, Dirty]	Suck	{A}
2	[A, Clean]	Right	[A, Clean]	Right	{A}
3	[B, Dirty]	Suck	[B, Dirty]	Suck	{A, B}
4	[B, Clean]	Left	[B, Clean]	NoOp	{A, B}
5	[A, Clean]	Right	[B, Clean]	NoOp	{A, B}
6	[B, Clean]	Left	[B, Clean]	NoOp	{A, B}

Note: Agent X moves from B to A at step 4, then from A to B at step 5 and from B to A at step 6. Agent Y remains in B after both squares are known clean.

(d) Total performance

Agent X cleans A and B once each: cleaning reward = 2 × 10 = 20. It makes four moves (Right, Left, Right, Left), costing 4 × 1 = 4. Total score X = 20 − 4 = 16.

Agent Y cleans A and B once each: cleaning reward = 20. It makes only one move (Right), costing 1. Total score Y = 20 − 1 = 19.

(e) Rationality and omniscience

Agent Y is rational under the stated performance measure because it uses an internal model/state to remember that both squares are clean and then chooses NoOp, avoiding unnecessary movement penalties. At step 4 both agents receive [B, Clean], but their programs differ: X is a simple-reflex agent and responds to the current percept alone by moving to the other square; Y is a model-based reflex agent and combines the current percept with its internal state {A, B}, so it chooses NoOp.

A rational agent chooses the action expected to maximize performance given its percept sequence and available knowledge. An omniscient agent would know the actual state and outcomes in advance, including facts not available to the rational agent. Rationality does not require omniscience.

QUESTION 2: FOUNDATIONS OF AI AND PROBLEM FORMULATION

(a) Four approaches to AI and historical milestones

System / idea	Approach to AI	Era / milestone
Turing Test proposed (1950)	Acting humanly	Early foundations of AI; Alan Turing's 1950 proposal for evaluating human-like conversational behavior.
General Problem Solver (Newell and Simon, 1957)	Thinking humanly	Early symbolic AI / cognitive simulation; aimed to model human problem-solving using means–ends analysis.
Aristotle's syllogisms and formal logic	Thinking rationally	Ancient foundations of logic; formal rules for valid reasoning.
Agent that maximizes expected performance	Acting rationally	Rational-agent approach; selecting actions to maximize expected performance.

(b) Water-jug problem formulation

Element	Formulation
State representation	(x, y), where x is the amount in jug X (0–4 L) and y is the amount in jug Y (0–3 L).
Initial state	(0, 0). Both jugs are empty.
Actions	Fill X; Fill Y; Empty X; Empty Y; Pour X into Y; Pour Y into X.
Transition model	Filling sets a jug to capacity; emptying sets its amount to 0; pouring transfers as much as possible until the source is empty or the target is full.
Goal test	x = 2, i.e. jug X contains exactly 2 litres.
Path cost	Each action costs 1; path cost is the number of actions.

(c) Successor states

From (0, 0), excluding unchanged states: Fill X → (4, 0); Fill Y → (0, 3). Emptying or pouring leaves the state unchanged.

From (4, 0), excluding unchanged states: Fill Y → (4, 3); Empty X → (0, 0); Pour X into Y → (1, 3). Empty Y and Pour Y into X leave the state unchanged.

(d) State-space size and shortest solution

There are 5 possible amounts in X (0, 1, 2, 3, 4) and 4 possible amounts in Y (0, 1, 2, 3), so the state space contains 5 × 4 = 20 states.

A shortest solution is: (0,0) → (0,3) → (3,0) → (3,3) → (4,2) → (0,2) → (2,0).

Actions: Fill Y; Pour Y into X; Fill Y; Pour Y into X; Empty X; Pour Y into X. Jug X ends with exactly 2 litres. The path cost is 6.

(e) Four phases of a problem-solving agent

1. Goal formulation: establish the goal that jug X contains exactly 2 litres. 2. Problem formulation: define states, initial state, legal actions, transition model, goal test and unit action costs. 3. Search: explore the state space to find a least-cost sequence reaching a state with x = 2. 4. Execution: perform the six actions in the plan. The agent can execute without new percepts because the task is deterministic and the initial state and action effects are known; this assumes the environment behaves as specified.

QUESTION 3: UNINFORMED SEARCH STRATEGIES

Graph edges and costs: S–A=1, S–B=4, A–C=2, A–D=5, B–D=1, C–G=6, D–G=2. Neighbours are expanded alphabetically. The goal test occurs when G is removed from the frontier.

(a) Breadth-first search (BFS)

Iteration	Dequeued	Queue after expansion (front left)	Explored set after expansion
1	S	[A, B]	{S}
2	A	[B, C, D]	{S, A}
3	B	[C, D]	{S, A, B}
4	C	[D, G]	{S, A, B, C}
5	D	[G]	{S, A, B, C, D}
6	G	[] — goal found	{S, A, B, C, D, G}

BFS path: S → A → C → G. Path cost = 1 + 2 + 6 = 9. Nodes removed from frontier = 6.

(b) Depth-first search (DFS)

The stack top is on the left. Children are pushed in reverse alphabetical order so the alphabetically first child is popped first.

Iteration	Node popped	Stack after pushing children (top left)
1	S	[A, B]
2	A	[C, D, B]
3	C	[G, D, B]
4	G	[D, B] — goal found

DFS path: S → A → C → G. Path cost = 1 + 2 + 6 = 9. Nodes removed from frontier = 4.

(c) Uniform-cost search (UCS)

Iteration	Node expanded	g(n)	Priority queue after expansion (lowest cost first)
1	S	0	[A:1(S), B:4(S)]
2	A	1	[C:3(A), B:4(S), D:6(A)]
3	C	3	[B:4(S), D:6(A), G:9(C)]
4	B	4	[D:5(B), G:9(C)] — D improved from 6 to 5
5	D	5	[G:7(D)] — G improved from 9 to 7
6	G	7	[] — goal found

UCS path: S → B → D → G. Path cost = 4 + 1 + 2 = 7. Nodes removed from frontier = 6.

(d) Comparison

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S → A → C → G	9	6	No — BFS minimizes number of edges, not weighted cost.
DFS	S → A → C → G	9	4	No — DFS is not generally cost-optimal.
UCS	S → B → D → G	7	6	Yes — with non-negative costs, UCS finds the least-cost path.

(e) Interpretation

UCS removes as many nodes as BFS in this graph (six) but finds a cheaper route because UCS orders the frontier by cumulative path cost, while BFS orders by depth. DFS reaches the goal after only four removals because it follows the A–C branch first, but this depends on traversal order and is not a guarantee of a cheapest path. An explored set prevents repeated expansion and cycles. At BFS iteration 3, B has neighbour S, which is already explored, and D, which is already in the queue; both must not be added again.

QUESTION 4: SEARCH WITH PARTIAL INFORMATION — BELIEF STATES

State mapping: S1=(L,D,D), S2=(R,D,D), S3=(L,D,C), S4=(R,D,C), S5=(L,C,D), S6=(R,C,D), S7=(L,C,C), S8=(R,C,C).

(a) Sensorless belief-state formulation

Element	Formulation
Initial belief state	{S1, S2, S3, S4, S5, S6, S7, S8}, because location and dirt status are initially unknown.

Actions	Left, Right, Suck.
Goal test	Every physical state in the belief state has both squares clean; the belief state must be a subset of {S7, S8}.
Path cost	Each action costs 1; total cost is the number of actions.

(b) Apply [Right, Suck, Left, Suck] without sensors

Step	Action	Belief state after action
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}
3	Left	{S3, S7}
4	Suck	{S7}

After the final Suck, the belief state is {S7}; both squares are clean in every state still considered possible.

(c) Belief-state prediction and percept update

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(d) Number of possible belief states

With 8 physical states, every subset of those states is a possible belief state, giving 2^8 = 256 belief states (including the empty subset). This is harder than searching over 8 physical states because the search space can grow exponentially with the number of physical states.

(e) Interpretation

The percept [Left, Dirty] removes every state in which the agent is not at Left or the left square is clean, leaving {S1, S3}. After Suck, the predicted belief is {S5, S7}; the [Left, Clean] percept is consistent with both, so neither is removed. After Right, the prediction is {S6, S8}; [Right, Dirty] removes S8 because its right square is clean, leaving {S6}. In the sensorless case, no percepts shrink the belief state; actions transform every possible state. The sensorless plan can still be rational because it guarantees that all states remaining possible after the plan have both squares clean, even without knowing the exact state.

FINAL ANSWER SUMMARY

Question	Key results
Q1	PEAS and environment classification completed; X = 16, Y = 19; model-based Agent Y is rational under the performance measure.
Q2	Turing Test—acting humanly; GPS—thinking humanly; Aristotle's logic—thinking rationally; maximizing expected performance—acting rationally. Water-jug state space = 20; shortest solution cost = 6.

Q3	BFS path S–A–C–G cost 9 (6 removals); DFS path S–A–C–G cost 9 (4 removals); UCS path S–B–D–G cost 7 (6 removals).
Q4	Sensorless sequence yields belief states {S2,S4,S6,S8}, {S4,S8}, {S3,S7}, {S7}. With percept updates: {S1,S3} → {S5,S7} → {S6,S8} → {S6}. Possible belief states = 256.